Statistical Analysis Plan (SAP)

Please complete all relevant sections. Instructions are in red and should be deleted when completing the SAP. The purpose of this template is to provide a general layout, but sections can be reformatted as you wish.

The creation of this template for the Duke BERD Methods Core was made possible by Grant Number UL1TR002553 from the National Center for Advancing Translational Sciences (NCATS) of the National Institutes of Health (NIH), and NIH Roadmap for Medical Research.

Title	Relative Effectiveness of Social Media, Dating Apps, and Information Search Sites in Promoting HIV Self-testing: Observational Cohort Study
CRU/Department/Division/Center	Department of Population and Public Health Sciences, University of Southern California
IRB Number	University of California, Los Angeles IRB #18-001580
Investigators:	
Primary Investigator	The corresponding author is Chrysovalantis Stafylis, MD, MPH (University of Southern California).
Collaborative Lead	Not reported in the manuscript.
Co-authors (if known)	Chrysovalantis Stafylis; Gabriella Vavala; Qiao Wang; Bethany McLeman; Shea M Lemley; Sean D Young; Haiyi Xie; Abigail G Matthews; Neal Oden; Leslie Revoredo; Dikla Shmueli-Blumberg; Emily G Hichborn; Erin McKelle; Landhing M Moran; Petra Jacobs; Lisa A Marsch; Jeffrey D Klausner.
Analysis Biostatistician(s)	Original analysis biostatistician is not reported in the manuscript. This SAP for replication author is Yuzhao (Clair) Tan
Biostatistics Supervisor	Not reported in the manuscript
Lead Biostatistician	Not reported in the manuscript
Subject Matter Expert	Not reported in the manuscript
Original Creation Date	Manuscript was submitted in 12 Dec 2021. This SAP created on 18 Jan 2026 for the portfolio course replication.
Version Date	Manuscript was published 23 Sep 2022; accepted 10 Aug 2022. The first SAP draft for portfolio course replication is on 18 Jan 2026.

	And this version is on 27, Jan 2026.
Project Folder Location	Not reported in the manuscript
Project Goal(s)	*Manuscript, abstract, presentation, etc.* Reproduce the published analyses for the course case study. Compare the relative effectiveness of online platform sites (social media, dating apps, and information search sites) in promoting HIV self-testing among young Black and Latinx MSM at increased HIV risk, using HIV self-test kit order rates as a proxy; and assess differences between participants who ordered vs. didn't order a kit.
Submission Deadline(s)	The deadline for the course SAP is 21 Jan 2026.
Effort Estimate (optional)	

Investigator Agreement	☒ All statistical analyses included in an abstract or manuscript should reflect the work of the biostatistician(s) listed on this SAP. No changes or additional analyses should be made to the results or findings without discussing with the project biostatistician(s). ☒ All biostatisticians on this SAP should be given sufficient time to review the full presentation, abstract, manuscript, or grant and be included as co-authors on any abstract or manuscript resulting from the analyses. ☒ If substantial additional analysis is necessary or the aims of the project change, a new SAP will need to be developed. ☒ Publications resulting from this SAP are supported in part by the Duke CTSA and must cite grant number UL1TR002553 and be submitted to PubMed Central. ☒I have reviewed the SAP and understand that any changes must be documented. *Acknowledged by:* Yuzhao (Clair) Tan *Date:* January 18, 2026
Activity Log	*This section should be used to track any amendments or changes made to the SAP. It should include detailed information about the changes and the date.* *This is where you should put any information about whether this is an entirely new SAP and an addendum was not appropriate. Here you should state why you had to write an entirely new SAP. Otherwise, include the changes as a 'change log' in the addendum section below. State reason and details if this is a new SAP for a project that has*

	undergone changes (i.e. dataset changed because the original one was not appropriate and a new SAP was created). *OPTIONAL: This section can also be used to track your project milestones/meetings/status if you find it helpful.* Version 1.0 (January 18, 2026): Initial SAP drafted to reproduce the published analysis plan for course case study replication. Version 2.0 (January 28, 2026): Final SAP draft to reproduce the published analysis plan for course case study replication.	
Acronyms	*MSM*	*Men who have sex with men*
	HIV	*Human immunodeficiency virus*
	PrEP	*Pre-exposure prophylaxis*
	IRB	*Institutional Review Board*
	NIDA	*National Institute on Drug Abuse*
	NIH	*National Institutes of Health*
	CTN	*Clinical Trials Network*
	TAPS	*Tobacco, Alcohol, Prescription medication, and other Substance use tool*
	SAP	*Statistical Analysis Plan*
	COVID-19	*Coronavirus disease 2019*
	CTR	*Click-through rate*
	CI	*Confidence interval*
	RR	*Rate ratio*
	VAS	*Visual analog scale*
	RCT	*Randomized controlled trial*

1 Study Overview

Background/Introduction:

HIV self testing offers a private and convenient option, but many people at higher risk may not access it without effective outreach. Online advertising can reach young Black and Latinx men who have sex with men through social media, dating apps, and search engines. Because platform audiences and user behavior differ, the impact of outreach may differ across platforms.

1.1 Study Aims

Primary aim

Aim 1 was to compare the daily HIV self test kit ordering rates across the recruitment sites within each recruitment wave.

Secondary aims

Aim 2 was to assess whether selected baseline measures differed between participants who ordered a kit within 60 days and participants who did not. The baseline measures were grouped into six secondary aims.

Secondary aim 1 was to compare substance use risk based on the Tobacco, Alcohol, Prescription medication, and other Substance use tool scores.

Secondary aim 2 was to compare stage of change for HIV testing based on the transtheoretical model categories.

Secondary aim 3 was to compare HIV related stigma measures.

Secondary aim 4 was to compare medical mistrust measures.

Secondary aim 5 was to compare attitudes toward HIV testing and beliefs about HIV treatment.

Secondary aim 6 was to compare PrEP awareness and PrEP opinions, including reported barriers and facilitators, and to describe PrEP related follow up outcomes at 14 and 60 days.

Exploratory aim
An exploratory aim was to describe advertising campaign performance metrics for each site, including impressions, clicks, click through rate, and total funds spent.

1.2 Study Hypotheses

This analysis is comparative (not purely descriptive). The published paper planned to compare platform types, but due to strong platform-by-wave interaction and large differences across sites within the same type, the final primary inference on within-wave site comparisons rather than pooling by platform type.

1.2.1 Primary Hypotheses

Within a given wave, at least one site has a different HIV self-test kit order rate (order per day) compared with another site in the same wave.

Null Hypothesis: the kit ordering rate was the same across all sites within a recruitment wave.

Alternative Hypothesis: at least one site had a different kit ordering rate within that wave. We also evaluated all pairwise site comparisons within each wave using rate ratios estimated from a Poisson regression model.

1.2.2 Secondary Hypotheses

Baseline measures of substance use, readiness to test (stage of change), and HIV-rated perceptions/attitudes differ between participants who ordered a kit within 60 days and those who did not.

Null Hypothesis: the mean or distribution of the measure was the same for participants who ordered a kit within 60 days and participants who did not order a kit.

Alternative Hypothesis: the mean or distribution differed between the two groups. All tests were two sided with an overall alpha of 0.05.

2 Study Population

2.1 Inclusion Criteria

- Men who have sex with men (MSM)
- Age 18-30 years
- Identify as Latinx or Black/African American (including multiracial and multiethnic individuals of these groups)
- Report either: condomless anal sex in the past 90 days, OR more than 1 male sex partner in the past 90 days

2.2 Exclusion Criteria

- HIV-positive status
- HIV testing in the past 90 days
- Currently taking PrEP or took PrEP at any time during the past 6 months prior to enrollment

Primary analysis exclusions:
Participants enrolled from Google and Facebook while Grindr was inactive early during Wave 1 (to ensure all three sites had equal opportunity to recruit during the compared period)

Wave 3 (COVID-19 early period) because no participants enrolled and no kits were ordered, making the model inestimable.

2.3 Data Acquisition

Fill in all relevant information:

Study design	Longitudinal observational cohort study with online recruitment via platform advertisements. Two planned advertising “waves” each containing one social media site, one dating app, and one information search site, run over the same period within the wave. Wave 1 had two phases due to an expected advertising shutdown on Grindr; the two Wave 1 phases were combined for analysis. Wave 3 was run during early COVID-19 (April 6-May 6, 2020) on different platforms but had zero enrollment/orders and was excluded from analysis
Data source/how the data were collected	Online advertising compaigns on: Social media (Facebook, Instagram); Dating apps (Grindr, Jask’D); Information search (Google, Bing). Baseline questionnaire; follow-up questionnaires at 14 and 60 days. Outcome tracking via the electronic ordering code used to order a free HIV self-test kit.

	Compaign performance metrics: impressions, clicks, click-through rate, and funds spent.
Contact information for team member responsible for data collection/acquisition	https://datashare.nida.nih.gov/study/nida-ctn-0083
Date or version (if downloaded, provide date)	Downloaded at Jan,16, 2026 from NIDA Data Share (CTN-0083). The study recruitment occurred between Jan an Sept 2020. In the manuscript, 271 participants were enrolled and 254 were included in the final analysis after exclusions.
Data transfer method and date	Downloaded via NIDA Data Share and saved locally for analysis.
Where dataset is stored	NIDA Data Share CTN-0083 dataset file in the local project folder.

Notes: Participant flow and analysis population: The manuscript reports 271 enrolled participants and a final analytic sample of 254 after exclusions. In the final analytic sample, 177 participants ordered at least one kit within 60 days and 77 did not order a kit. The primary analysis uses a per protocol style analytic set, excluding participants enrolled from Google and Facebook during days when Grindr was inactive early in Wave 1, and excluding Wave 3.

Description:
De-identified NIH/NIDA CTN-0083 dataset downloaded from NIDA Data Share. The analysis dataset includes participant-level baseline and follow-up survey responses, recruitment source (site and wave), timestamps for enrollment and kit ordering, and campaign performance metrics (impressions, clicks, costs).
Data dictionary: NIDA Data Share CTN-0083 codebook/data dictionary (downloaded alongside CTN_FINAL.csv).

3 Outcomes, Exposures, and Additional Variables of Interest

A. *If you choose not to use the table format below, make sure all the information outlined below is still included.*

3.1 Primary Outcome(s)

You should only have 1-2 primary outcomes and the rest should be secondary.

Outcome	Description	Variables and Source	Specifications
Daily HIV self-test kit order rate	Number of home HIV self-test kits ordered per day during each advertising wave/site	Order counts from ordering system (via unique e-code), and advertising period days	e.g., how is it coded? Order rate = (number of orders)/(number of advertising days). Computed separately for each site and wave. Primary comparisons are within-wave across the 3 sites.

3.2 Secondary Outcome(s)

Outcome	Description	Variables and Source	Specifications
Order a kit with in 60 days	Whether a participant ordered a self-test kit within 60 days of the code being emailed	Order tracking and enrollment date	e.g., how is it coded? Binary: ordered vs. not ordered. Participants who didn't order within 60 days are classified as "not ordered".
Self-test use and results (follow-up)	Self-reported use and results of the ordered kit	Follow-up surveys (14 and 60 days)	Descriptive reporting; not the primary effectiveness metric.
PrEP-related follow-up outcomes	Visiting a provider to discuss PrEP/ starting PrEP after testing	Follow up surveys at 14 days and 60 days.	Reported descriptively. We will summarize counts and percentages among follow up respondents.

3.3 Additional Variables of Interest

This is optional but generally this would be all your covariates of interest. If there are any variables that need special calculations etc. include them here or in a data dictionary that is an appendix to this SAP which you can reference here.

Variable	Description	Variables and Source	Specifications
Recruitment site and wave	Which platform site recruited the participant; wave 1 vs wave 2	Advertising records / dataset variable	Categorical: Facebook, Grindr, Google, Instagram, Jack'D, Bing; Wave indicator.
Platform type	Social media vs dating app vs search	Derived from site	Used descriptively; not pooled for primary inference due to interaction issues.
Substance use (TAPS)	Tobacco, Alcohol, Prescription medication, other substance	Baseline survey; TAPS scoring	For each substance: score=1 → "problem use"; score ≥2 → "high-risk use."
Stage of change for HIV testing	Readiness to test (transtheoretical model)	Baseline survey	Ordinal categories (e.g., determination, maintenance, etc.).
HIV stigma, medical mistrust	Psychosocial scales	Baseline survey	Scale totals or item-level summaries per

			instrument guidance.
Attitudes toward HIV treatment	10-item visual analog scale questionnaire	Baseline survey	Treat as continuous; report mean (SD) per item.
PrEP opinions/barriers/facilitators	Likert-scale items	Follow-up survey	5-point Likert; compare ordered vs not ordered descriptively.
Demographics and sexual risk behaviors	Age, race/ethnicity, condomless receptive anal sex, number of partners, etc.	Baseline survey	Standard descriptive summaries; selected comparisons for ordered vs not ordered.
Selected baseline stigma and knowledge items	Three baseline survey items used as targeted secondary checks to reproduce reported p values.	Baseline survey item responses.	Likert scale responses treated as ordinal. We will compare ordered within 60 days versus not ordered using a Wilcoxon rank sum test. Targeted items include People would leave if I had HIV, New HIV AIDS treatments can eradicate the virus, and Could not be friends with someone with HIV AIDS.

Missing Data

We will describe missingness for key variables (especially follow-up outcomes at 14 and 60 days and any baseline items). For primary order-rate analyses, missingness is not expected for site-level order counts on active advertising days. For participant-level comparisons, we will use complete-case analyses consistent with the manuscript and report the number of participants with non-missing data for each variable.

4. Statistical Analysis Plan

Every analysis step should refer back to an aim/hypothesis/objective. You can include just one deadline for the project, but if there are multiple major stopping points, you should include individual deadlines for each milestone. Good practice is to number each item here and reference them in your code. If you *have to wait for data, you should include a disclaimer that X analysis cannot be completed until the data is collected and deadlines cannot be determined until you receive the data.*

For more details on what to include, see the SAP Checklist document.

1. Analyses will follow the published paper's approach.
2. Statistical significance will be assessed using two sided tests with an overall alpha of 0.05.
3. Multiple comparisons for primary site contrasts will be adjusted using the Hochberg procedure.
4. Descriptive statistics:
   a) Continuous variables: median (25th, 75^{th} percentiles) and/or mean (SD), as appropriate.
   b) Categorical variables: counts and percentages.
5. Software:
   a) The original paper used SAS 9.4
   b) For replication, analyses may be conducted in R (version documented in the reproducibility log) with scripts that generate the same tables/figures.

4.1 Demographic and Clinical Characteristics ("Table 1")

1. Created a baseline characteristics table for the final analytic sample (paper's per-protocol analysis set; N=254).
2. Summarize: Age; Race/ethnicity; Key sexual behavior measures (e.g. number of partners in past 90 days; condomless receptive anal sex in past 90 days); HIV testing history and timing since last test; Any other baseline measures needed for secondary analyses.
3. Also present baseline characteristics by kit ordering status (ordered within 60 days vs. not ordered), consistent with the paper's secondary analyses.

4.2 Analyses Plan for Aim 1

Outcome: daily order rate (orders per day) for each site during active advertising days within each wave.

Data preparation step:

1. Define the analytic sample:
   - Exclude participants enrolled from Google/Facebook when Grindr was inactive early in Wave 1.
   - Exclude Wave 3 observations (no enrollment/orders).
   - Combine the two Wave 1 advertising periods into a single Wave 1 dataset (as in paper).
2. For each site within each wave, compute:
   - number of orders
   - number of advertising days
   - order rate = orders / advertising days

Primary statistical comparisons step:

3. Fit Poisson regression models for count outcomes with exposure time (advertising days) to estimate and compare order rates. Use post-hoc contrasts to compare sites within the same wave. The planned contrasts are:

- Wave 1: Facebook vs Grindr; Facebook vs Google; Grindr vs Google
- Wave 2: Instagram vs Jack'D; Instagram vs Bing; Jack'D vs Bing

4. Adjust the set of pairwise contrast p-values using the Hochberg method to control for multiple testing.

5. Report for each contrast:
    - estimated rate ratio (or equivalent contrast measure)
    - 95% confidence interval
    - Hochberg-adjusted p-value
6. Interpretation:
    - Identify which site(s) show higher order rates within each wave.
    - Avoid pooling by platform type in the primary inference, consistent with the paper, due to platform-by-wave interaction and large within-type variability.

4.3 Analyses Plan for Aim 2

Outcome: ordered within 60 days (yes/no).

1. Define the binary outcome:
   "Ordered" if a participant ordered a kit within 60 days of receiving the e-code.
   "Not ordered" otherwise.

2. Compare baseline measures between ordered vs not ordered participants:
   Continuous variables: Student t test (if appropriate) or nonparametric alternative as needed.
   Categorical variables: Fisher exact test.
   Likert-scale responses: Wilcoxon rank-sum test.
3. Report:
   Group summaries (ordered vs not ordered)
   p-values for differences
4. These analyses are primarily associative and descriptive; they do not establish causality due to observational design.

5 Missing Data

5.1 Primary outcome (orders and advertising days): no missing data are expected because ordering is tracked via the electronic ordering system and ad periods are known.

5.2 Survey questions that participants could skip:
- Skipped responses will be coded as missing;
- Missing responses will be excluded from the denominator when calculating frequencies for those items, consistent with the paper's approach.

5.3 Follow-up survey nonresponse:
- Describe the amount of missing follow-up data.
- Follow the paper's main approach (complete-case for each item), and clearly state denominators.

6 Sensitivity Analyses

Conduct the following three sensitivity analyses using the same Poisson regression and contrast framework as the primary analysis:

1. Any-time ordering sensitivity: classify orders as occurring at any time during the study (not limited to the 60-day window) and re-run primary comparisons.
2. Wave 1 two-phase sensitivity: address the two-phase nature of Wave 1 by evaluating whether treating phases separately vs combined materially changes inference.

3. COVID-19 impact sensitivity: assess the impact of the COVID-19 period; Wave 3 is excluded from the main analysis because it had zero enrollment/orders, but its context and implications will be discussed as a limitation.

7 Limitations

All design and analysis limitations (this will grow as you do the study and should be included in the report)

1. Observational design with self-selection into the cohort; differences across sites may reflect differences in users rather than platform effects.
2. Advertising was organized in waves; time-varying external events (including the COVID-19 pandemic) may confound comparisons, and between-wave comparisons are limited.
3. The primary outcome is ordering a kit, which is a proxy for testing behavior and does not guarantee test use.
4. Many secondary outcomes are self-reported and may have missing data or reporting bias.
5. Platform policies and ad delivery algorithms can change over time, which may limit generalizability.
6. The sample was limited to young Black and or Latinx men who have sex with men in selected states and the District of Columbia, and people had to be willing to respond to online ads, so the findings may not generalize to other groups or settings.

8 Addendum for Additional Analyses

Not applicable for the initial SAP. Any unplanned analyses conducted after the SAP is finalized will be documented here with date, purpose, and methods.

9 Appendix

Table 1 shell: Baseline characteristics overall and by ordered within 60 days (n, %, mean/SD, median/IQR).
Table 2 shell: Orders, advertising days, and order rates by site and wave; plus pairwise contrast results with Hochberg-adjusted p-values.
Figure shell: Bar/point plot of order rates by site (separately for Wave 1 and Wave 2) with confidence intervals.

10 References

1. Stafylis, C., et al. (2022). Relative effectiveness of social media, dating apps, and information search sites in promoting HIV self testing: Observational cohort study. JMIR Formative Research, 6(9), e35648. https://doi.org/10.2196/35648
2. Hochberg multiple comparison procedure (for adjusted p-values).
3. R session info (replication).

Statistical Analysis Plan Checklist

Below you will find a checklist of recommended items to include in a statistical analysis plan. Some of these are specific to clinical trials (based on this JAMA paper) and some are other are specific to observational studies (based on STROBE/RECORD guidelines), so every item will not be necessary for every project. The biostatistician should start with the SAP template above and add in necessary information from the checklist. Item numbers that are starred (*) are not explicitly included in the SAP template and should be added by the author if relevant to the project. This checklist was developed using the CONSORT 2010 Checklist.

Section/Topic	Item #	Description	Included (Yes/No/NA)
Administrative Information			
Study Information	1a	Descriptive title that matches the protocol, with SAP either as a forerunner or subtitle	______
	1b	Trial registration number, protocol version number, and/or IRB number.	______
	1c	CRU/Department/Division/Center/other collaborative unit that the study falls under	______
Roles and responsibility	2a	Listing of principal investigators, clinical leads, and co-authors (if known)	______
	2b	Name and affiliation of SAP author(s)	______
	2c	Names, affiliations, and roles of other SAP contributors (e.g. senior statistician)	______
SAP Information	3	SAP version number, with date of current version and original creation date	______
Project Information	4a	Project folder location	______
	4b	Project goals (e.g. manuscript, abstract, presentation, etc.)	______
	4c	Project deadlines (of listed goals)	______
	4d	Effort estimate	______
Investigator Agreement			
Investigator Agreement	5	Confirmation that BERD Method Core's collaborative process has been reviewed, that all statistical analyses included in an abstract or manuscript should reflect the SAP, no changes should be made to the SAP without discussing with the SAP author, all biostatisticians on the SAP are co-authors on the manuscript, and that publications resulting from the SAP must cite grant number UL1TR002553 and be submitted to PubMed Central	______
Signatures	6	Signatures of SAP author, senior statistician, and principal investigator(s)	______
Activity Log			
SAP revisions	7a	SAP revision history with dates	______
	7b	Justification for each SAP revision	______
	7c*	Timing of SAP revision in relation to any interim analyses or submissions	______

[1]Turner L, Shamseer L, Altman DG, et al. Consolidated standards of reporting trials (CONSORT) and the completeness of reporting of randomised controlled trials (RCTs) published in medical journals. *Cochrane Database Syst Rev*. 2012;11:MR000030. Published 2012 Nov 14. doi:10.1002/14651858.MR000030.pub2

Study Overview			
Background and introduction	8	Synopsis of scientific background and rationale for the study	______
Aims and Hypotheses	9a	List of all scientific aims/objectives of the study, with specifications of primary, secondary, etc.	______
	9b	List of all statistical hypotheses (corresponding to the scientific aims), with specifications of primary, secondary, etc.	______
Variables of Interest	10a	List of all outcome/endpoint variables, with a description of their coding/units, timing, and source, corresponding to the statistical hypotheses. If any variables are defined using ICD or CPT codes, list them out.	______
	10b	List of all exposure variables, with a description of their coding/units, timing, and source, corresponding to the statistical hypotheses. If any variables are defined using ICD or CPT codes, list them out.	______
	10c	List of any additional variables of interest (e.g. covariates, potential confounders, effect modifiers, etc.) in the analysis	______
	10d*	Location of data dictionary (or provided as an appendix)	______
	10e*	Report category boundaries if continuous variables are collapsed into categories, and describe any other relevant data transformations	______
Causal Graph	11*	May be helpful to include a DAG or other graph/diagram that describes the way the variables of interest are presumed to relate to each other	______
Study Methods			
Study Plan and Design	12a	Description of the study design (e.g. parallel group randomized trial, case-control study, cohort study, etc.)	______
	12b*	Study setting, location, and relevant dates (e.g. periods of enrolment, exposure, follow-up, and collection)	______
	12c*	Description of intervention or exposure groups, with allocation ratios, and details of any matching criteria	______
	12d*	Details on randomization (e.g. stratification factors) and blinding procedures	______
	12e	List of eligibility and/or inclusion/exclusion criteria	______
	12f*	Description of screening/enrolment/recruitment processes	______
	12g*	Description of patient flow (e.g. CONSORT diagram)	______
	12h*	Description of analysis population (e.g. intention to treat, per protocol, etc.)	______
	12i*	Definitions of adherence/compliance, protocol deviations, loss-to-follow-up, adverse events, etc.	______
	12j*	Time points at which outcomes are measured	______

	12k*	Timing of final analyses (are all outcomes analysed collectively, or will short-term outcomes be analysed separately from long-term outcomes, etc.)	
Sample Size	13a*	Sample size calculation or justification (either provided in full or summarized, with link to original source)	
	13b*	Description of pre-planned subgroup analyses, power for these analyses, and planned multiple comparison adjustment procedures	
Interim Analyses	14a*	Description of what interim analyses will be conducted at which time points, and what methods used to adjust significance levels due to the interim analysis	
	14b*	Details of any guidelines (e.g. safety, futility) for stopping the study early	
	14c*	Details of any changes to trial design due to interim analyses (e.g. enrolling more patients)	
Data	15a	Description of data collection/acquisition process, with contact information for team member responsible	
	15b	Description of data flow/transfer from primary data collection through to creation of final analysis dataset	
	15c	Data transfer method and date	
	15d	Folder location where datasets are stored	
	15e*	Description of any additional data management, quality control, or processing undertaken	
	15f*	If any data are extracted from a database, a description of the database and the query used for the extraction, and whether/how it was merged with any data from outside that database. If the study involved linkage of databases, consider use of a flow diagram to demonstrate the data linkage process, including the number of individuals with linked data at each stage.	
	15f*	Description of any other data sources incorporated in the analysis	
Missing Data	16a*	Description of sources and magnitudes of missing data	
	16b*	Description of how missing data patterns will be presented/summarized (may be helpful to have a table shell or draft CONSORT-style diagram)	
	16c*	Description of contingency plans for handling missing data in analysis	
Simulations	17a*	If conducting a simulation, a description of the purpose of the simulation and its design (e.g. fully factorial, partially factorial, grid search, etc.)	
	17b*	Define the fixed and variable factors or parameters in the simulation, the estimands/targets of the simulation, and the performance measures to be estimated (with justifications of their relevance to the estimands/targets)	
	17c*	Description of the tabular and graphical presentations of simulation results and their interpretation	

Statistical Analysis Plan

Statistical Significance	18a*	Hypothesis testing framework (e.g. superiority, equivalence, non-inferiority), or description of alternative analytic framework (e.g. evaluation of a posterior in a Bayesian analysis, etc.)	
	18b*	Level of significance for primary hypotheses, including a description and rationale for any multiple comparisons adjustment or Type I error control procedures	
	18c*	Description of any decision-making rules based on confidence intervals, credible intervals, prediction intervals, Bayes' factors, or other alternative inferential methods	
	18d*	Description of how the results of any hypothesis tests (or alternative inferential methods) will be interpreted with respect to both the statistical hypotheses and scientific aims/objectives of the study	
Descriptive Statistics	19a*	List of characteristics (e.g. demographic, clinical) to be summarized descriptively (e.g. "Table 1")	
	19b*	Description of how these characteristics will be summarized descriptively (e.g. means/medians vs. N (%), tabular displays, graphical displays, etc.)	
	19c*	Summarize follow-up time (e.g. average and total amount) and number of events	
Analysis Methods	20a	For each aim/hypothesis (see items 9a/9b), a description of what analysis method will be used and how the results from this method will be reported and interpreted	
	20b*	Description of any transformations, standardizations, covariate or confounder adjustments, weighting, or stratification methods to be used and why.	
	20c*	For each analytic method proposed, a description of the assumptions of that method and what processes will be used to evaluate whether or not those assumptions hold	
	20d*	Details of contingency plans/alternative methods to be used if the assumptions are found not to hold	
	20e*	In the case of non-standard test statistics, formulas provided for the test statistic with a description of the mathematical null hypothesis, how significance is determined, and how the test statistic is interpreted	
	20f*	In the case of regression models, formulas provided for the full model with a description of which parameters are to be used, how they will be interpreted, how confidence intervals will be constructed, etc.	
	20g*	In the case of survey, hierarchical/nested, or clustered data, a description of what methods will be used to adjust for the data structure and why (e.g. if using a GEE, describing which correlation structure and why it was chosen, etc.)	
	20h*	For non-continuous outcomes, clearly explain the effect used (e.g. risk difference, risk ratio, odds ratio, etc.), whether it is relative or absolute, and justify why that was chosen as the effect measure of interest	
	20i*	Documentation of any non-standard methods used (e.g. using alternative degree of freedom calculation methods, using a non-canonical link function, etc.)	

	20j*	Description of any limitations, sources of bias, internal/external validity, and other relevant discussions concerning the interpretation and generalizability of the design or methods used	
Additional Analysis Methods	21a*	Description of any pre-planned sensitivity analyses and how they will be interpreted	
	21b*	Description of pre-planned subgroup analyses, power for these analyses, and planned multiple comparison adjustment procedures	
	21c*	Description of any additional post-hoc calculations or analyses (e.g. evaluating interaction/modification effects, calculating mediation or local average treatment effects, evaluation of AUROC curves, etc.)	
	21d*	If conducting any bootstrap analyses, a description of the sampling algorithm and number of iterations used	
	21e*	If conducting any cross-validation procedures, a description of how the cross-validation is conducted (e.g. leave-one-out, train/validation/test, etc.)	
Exploratory Analyses	22a*	Description and justification for any pre-planned exploratory analyses and what methods will be used to conduct them	
	22b*	Framework for conducting any unplanned exploratory analyses and how they will be integrated into the planned analysis	
Software	23*	List of statistical software (along with version numbers) to be used for each phase of the analysis; in the case of R or Stata, additionally list any requisite installed packages and their version numbers	
Other	24*	Description of any additional planned analyses of the data (e.g. a safety analysis looking at adverse event rates for a Data Safety Monitoring Board, etc.)	
Tables and Figures			
Table Shells	25*	Example tables related to any of the conducted analyses; if possible including any available preliminary data	
Example Figures	26*	Example figures related to any of the conducted analyses; if possible including any available preliminary data.	
References			
References	27a	References for any non-standard statistical methods used	
	27b	References (and locations) for any relevant protocols, standard operating procedures, or other documents cited in the SAP	
Additional Information			
Appendices	28*	If necessary, appendices may be included (e.g. a full data dictionary, a copy of a Case Report Form, etc.)	
Addendums	29*	Any additional analyses conducted that were not included in the SAP should be documented in an addendum, describing the purpose of the additional analysis, when it was conducted, and by whom	